

**Installation Instructions for the Community Satellite Processing Package (CSPP) VIIRS, ATMS,
CrIS and OMPS SDR Version 4.1.1 Software for Suomi-NPP, NOAA-20 and NOAA-21**

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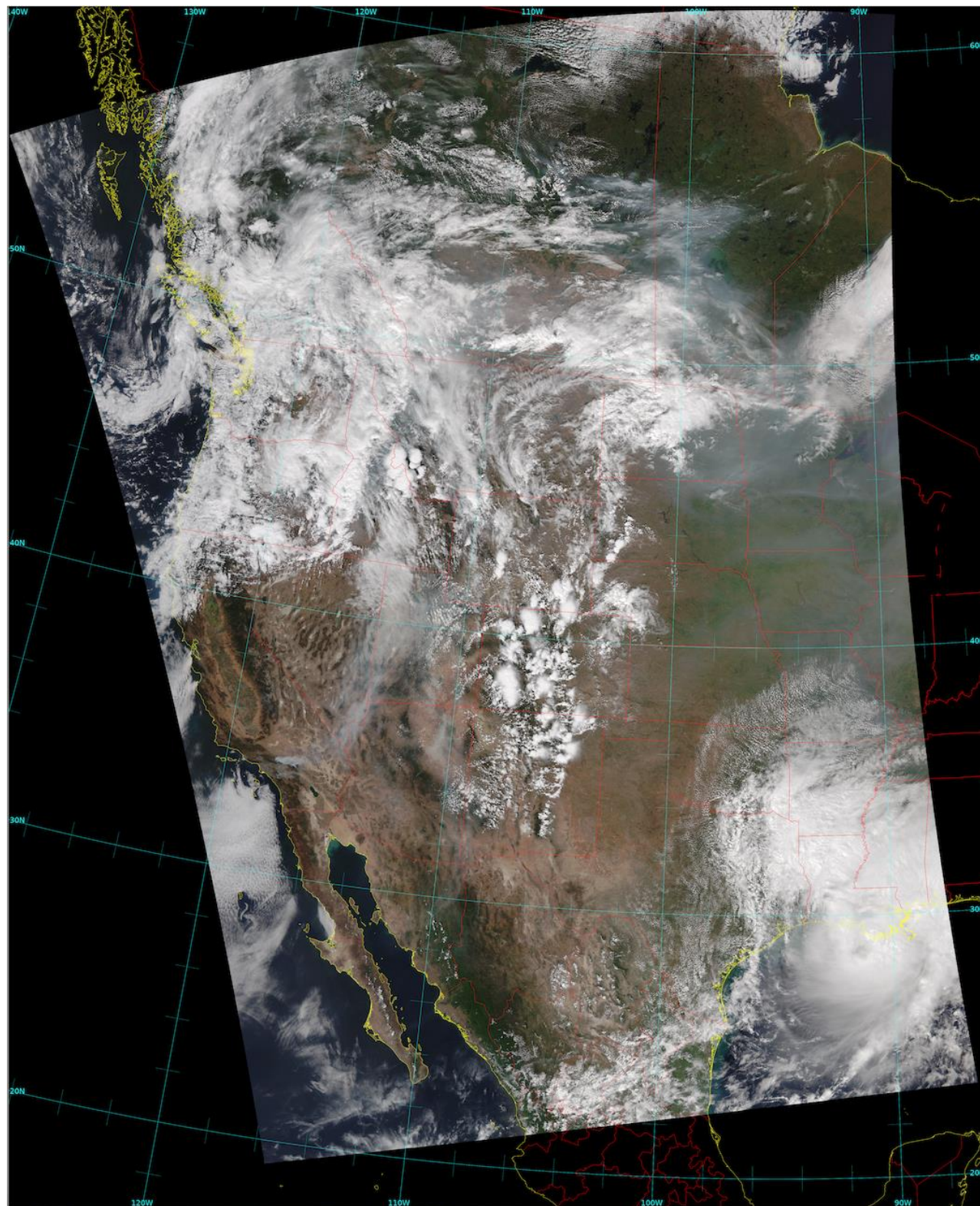


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Section 1: Introduction

1.1 Overview

This guide provides instructions for the installation and operation of the Community Satellite Processing Package (CSPP) software. The CSPP transforms direct broadcast data from Suomi-NPP, NOAA-20, and NOAA-21 VIIRS, ATMS, CrIS, and OMPS Raw Data Records (RDR) into Sensor Data Records (SDRs) on Intel Linux computers. It includes binary executable files, supporting static data files, and input/output files for installation verification.

The CSPP SDR software is based on the Algorithm Development Library (ADL) software developed by Raytheon for the NOAA/NASA Joint Polar Satellite System (JPSS) Project. The ADL allows JPSS satellite instrument processing algorithms to run unmodified in a Linux environment. The Space Science and Engineering Center (SSEC) has packaged these algorithms for real-time direct broadcast mode via the Linux command line, retaining the original processing software, algorithms, and data formats. Note that source code is not included; it's available separately from the ADL distribution. The output files generated are identical in name, format (HDF5), and structure to those from the NOAA/NESDIS CLASS archive. For SDR file format descriptions, refer to the JPSS Common Data Format Control Book (CDFCB) External Vol. III at:

<https://www.jpss.noaa.gov/sciencedocs.html>

The CSPP SDR software is available through the CSPP website:

<https://cimss.ssec.wisc.edu/cspp/>

Download software, test data, and documentation from the site. Please use the 'Contact Us' form on the website to submit any questions or comments about CSPP.

1.2 What's New in CSPP SDR Version 4.1.1

- Updated ATMS executable in the form of a patch that affects ATMS SDR geolocations for all JPSS missions (NOAA-20, NOAA-21 and S-NPP). Instructions are updated in the [Installation Section 3](#) for both new installations and applying the patch to an existing CSPP SDR Version 4.1 installation.

1.3 What's New in CSPP SDR Version 4.1

- Updated code base: ADL Block 2.3 Mx10 (ADL 5.3_I2.3.10.00).
- Removed RDR Spacecraft Diary Filtering. See [Section 2.2.4](#).
- New Look-Up-Table (LUT) external distribution site:
 - JPSS_REMOTE_LUT_DIR=http://jpssdb.ssec.wisc.edu/cspp_luts_v7_0/
- Target Operating System is now Rocky Linux 8.

1.3 System Requirements

System requirements for the CSPP SDR Version 4.1 software are as follows:

- 64-bit Intel or AMD CPU with 8 cores (minimum);
- 32 GB RAM (minimum), plus an additional 4GB of RAM for every core that is used for CSPP VIIRS SDR processing;
- Rocky Linux 8.10 (the software has also been tested on Rocky Linux 9.4);
- 150 GB disk space (minimum);
- Internet connection (for downloading ancillary data).

Note: For routine real-time processing of S-NPP, NOAA-20, and NOAA-21 HRD, a Linux server running Rocky Linux 8 with 64 cores, 256 GB RAM, and 8 TB hard drive space is recommended.

1.4 Input Data Requirements

VIIRS, CrIS, ATMS and OMPS Raw Data Record (RDR) files in HDF5 format are required as input to the CSPP SDR Version 4.1 processing software. RT-STPS version 7.0 or creates RDR files that can be ingested by the CSPP SDR processing software.

The following are examples of valid RDR input file names (NOAA-21 Examples):

VIIRS - Visible Infrared Imaging Radiometer Suite

RNSCA-RVIIRS_j02_d20240911_t2050321_e2056135_b00001_c20240911205504076000_dr1u_ops.h5

ATMS - Advanced Technology Microwave Sounder

RATMS-RNSCA_j02_d20240911_t2051156_e2054596_b00001_c20240911205504398000_dr1u_ops.h5

CrIS - Cross-track Infrared Sounder

RCRIS-RNSCA_j02_d20240911_t2051476_e2052196_b09528_c20240911211013196008_dr1u_ops.h5

OMPS - Ozone Mapping and Profiler Suite

Nadir Mapper

RNSCA-ROTCS_j02_d20240911_t2051239_e2055084_b00001_c20240911205504478000_dr1u_ops.h5

Nadir Profiler

RNSCA-RONPS_j02_d20240911_t2051239_e2055084_b00001_c20240911205504406000_dr1u_ops.h5

1.5 Disclaimer

Original scripts and automation included as part of this package are distributed under the GNU GENERAL PUBLIC LICENSE agreement version 3. Binary executable files included as part of this software package are copyrighted and licensed by their respective organizations, and distributed consistent with their licensing terms.

The University of Wisconsin-Madison Space Science and Engineering Center (SSEC) makes no warranty of any kind with regard to the CSPP software or any accompanying documentation, including but not limited to the implied warranties of merchantability and fitness for a particular purpose. SSEC does not indemnify any infringement of copyright, patent, or trademark through the use or modification of this software.

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Section 2: Implementation and Configuration

2.1 Overview

This software package contains the JPSS Algorithm Development Library (ADL) version ADL 5.3_I2.3.10.00 SDR algorithm codebase for VIIRS, ATMS, CrIS and OMPS instruments bundled together as a standalone processing system. The SDR algorithms correspond to the Block 2.3 Mx10 version and are compiled for Intel Linux 64-bit (“x86_64”) systems compatible with Rocky Linux 8.10. The software reads RDRs and writes SDRs in a HDF5 format compatible with the JPSS Common Data Format Control Books (CDFCBs).

The CSPP SDR Version 4.1.1 patch implements the ATMS Beam Alignment Correction Rotation Matrix update that went into the JPSS global data processing system at 15 UTC on December 15, 2025. This update included a code change for all ATMS instruments (SNPP, NOAA-20, NOAA-21). The application of the patch improves the geolocation for all instruments.

2.2 CSPP SDR Version 4.1 Implementation

- **Replacement Release:** This release replaces existing installations of CSPP SDR (it is not an update). The CSPP SDR v4.1 executables and static tarfiles must be used for this release. The tarfiles from previous releases cannot be reused.
- **Look-Up-Table (LUT) Files:** To limit the size of individual release tarballs, the Look-Up-Table (LUT) files are separated based upon satellite.
- **Software Updates:** This release is based upon ADL 5.3_I2.3.10.00, and includes software changes introduced as a result of S-NPP, NOAA-20 and NOAA -21 Calibration/Validation efforts.

2.2.1 Processing Historical Data Sets with CSPP SDR Version 4.1

CSPP SDR v4.1 supports creating VIIRS, CrIS, ATMS and OMPS SDRs from historical NOAA-21, NOAA-20 and S-NPP RDRs beginning with data collected on 23 May 2024.

To process historical data, first download the historical LUTs using the `When processing historical data, you must first make sure to download the historical Look-Up-Tables (LUTs) by using use the -t option with sdr_luts.sh. For example:`

```
sdr_luts.sh -t 20240523
```

For more information, see [Section 3.2](#).

2.2.2 Cross-Granule Context Requirements

All NOAA-21, NOAA-20 and S-NPP instruments (VIIRS, CrIS, ATMS and OMPS) have cross-granule requirements that are used for proper calibration. These requirements vary by instrument and can affect the output granule yield.

VIIRS: The first and last granules of an overpass are used for context and are not processed by the SDR algorithm. Although the VIIRS processing will produce output for a single granule, there will be a line of fill values at the beginning and end of the granule. If these individual granules are then aggregated, seams will be evident along the granule boundaries. Optimally, VIIRS uses a context of 1 RDR granule before and 1 RDR granule after the granule being processed (VIIRS granules are approximately 86 seconds long).

CrIS: The context requirements for CrIS SDR are 4 RDR granules before and after the granule being processed. A sequence of 9 consecutive RDR granules is required to process a single granule to SDR (CrIS granules are 32 seconds long).

ATMS: The context requirements for ATMS SDR are 1 RDR granule before after the granule being processed. A sequence of 3 consecutive RDR granules is required to process a single granule to SDR (ATMS granules are 32 seconds long).

OMPS: The context requirements for OMPS Nadir Mapper and Nadir Profiler SDR are 1 RDR granule before and after the granule being processed. A sequence of 3 consecutive RDR granules is required to process a single granule to SDR.

Note that the yield of VIIRS, CrIS, ATMS and OMPS SDR granules may be reduced if there are incomplete RDR input granules.

2.2.3 Direct Broadcast Data Truncation

To accommodate the transmission of CrIS Full Spectral Resolution (FSR) observations, S-NPP and NOAA-20 VIIRS M-Band 7 is excluded from the High Rate Data (HRD) direct broadcast transmission. Additionally, the S-NPP satellite does not include CrIS Field-of-View (FOV) number 4 (beginning 4 December 2014) and number 6 (beginning 8 August 2015) from HRD. The un-truncated data sets are available from the NOAA CLASS archive. NOAA-21 HRD transmission includes the full suite of un-truncated CrIS and VIIRS data.

As in the previous CSPP SDR version, the CSPP V4.1 software handles this S-NPP and NOAA-20 data deletion with the following approach:

CrIS SDR

- S-NPP CrIS FSR SDRs contain fill values for missing FOVs 4 and 6.

VIIRS SDR

- S-NPP and NOAA-20 downlink of CrIS FSR data comes at the expense of the downlink of VIIRS M-Band 7 (M7) (0.865 micron). CSPP SDR v4.1 reconstructs VIIRS M7 using data from VIIRS I-Band 2 (I2) if the SDR software detects that the M7 data set contains fill values, and the I2 data contains valid data.
- The reconstructed M7 data is very similar (but not identical) to the M7 Stored Mission Data (SMD). The algorithm used by CSPP SDR v4.1 for reconstruction is very simple. It computes the average of the overlapping I2 FOVs to create the M7 replacement FOV. If any of the I2 data values that go into producing one M7 FOV are missing, a missing value is stored in the M7 output FOV.

- M7 is a dual gain channel, so differences in the reconstructed array may be most noticeable in dark (small radiance/reflectance) scenes. Downstream products that use M7 reflectances will be affected, including the VIIRS cloud mask product.
- M7 reconstruction requires setting quality flags for the reconstructed VIIRS M7 data. Since the format of the quality flags for both I-bands and M-Bands are identical, combining the four I2 data quality values into the final M7 quality flag is done using the logical “OR” function. If any of the bits that go into a quality flag value is set (1) for any of the I2 pixels, then the reconstructed M7 quality flag bit will also be set. Although this may not perfectly preserve all quality information stored in the I2 quality flag arrays for all cases, it will be enough to alert users of potential problems with a given Field-of-View.

ATMS and OMPS SDR

- ATMS and OMPS instruments are not affected by the HRD downlink truncation.

2.2.4 Direct Broadcast Data Dropouts and CSPP Filtering

CSPP SDR Version 4.1 no longer includes Spacecraft Diary RDR size filtering as was done in SDR Version 4.0. This practice was originally implemented to mitigate the impact of direct broadcast data dropouts on instrument calibration and geolocation. However, improved handling of missing data in the ADL software, and fluid RDR Diary standard sizes led to the removal of this filtering.

Science RDR size filtering continues to be implemented by default for CrIS and ATMS instruments.

2.2.5 S-NPP CrIS Short Wave/Long Wave Configuration

An anomaly on the S-NPP CrIS instrument on 21 May 2021 resulted in the loss of the Long Wave Band. Consequently, the JPSS project switched to a Short Wave/Long Wave configuration using Side 1 electronics, making the Mid Wave Band unavailable from S-NPP CrIS instrument.

Section 3: Installation of VIIRS, ATMS, CrIS and OMPS SDR

3.1 New CSPP SDR Version 4.1.1 Installation

There are two options for downloading and installing the CSPP SDR Version 4.1.1 software:

1. **Automatic Installation:** Use a Borne Shell (Bash) script to automatically download, unpack, install, and prepare the 9 tarballs that comprise the Version 4.1.1 release, including the 4.1.1 patch.
2. **Manual Installation:** Users can manually install the new software, similar to legacy versions.

3.1.1 Script Based Installation

To install the CSPP SDR Version 4.1.1 software using a bash script, follow the instructions below.

1. Download the CSPP Version 4.1.1 installation script into your \$HOME directory:
`cd $HOME`
`wget https://bin.ssec.wisc.edu/pub/CSPP/support/cspp\_sdr\_4.1.1\_install.bash`
2. Ensure that the script is executable on your system:
`chmod guo+x cspp_sdr_4.1.1_install.bash`
3. Change to the installation directory on your local machine:
`cd <local installation directory>`
4. Execute the script from the installation directory:
`$HOME/cspp_sdr_4.1.1_install.bash`

Executing the script performs the following tasks:

- Downloads 10 tarballs and the sha1sum into the <<installation_directory>/tarfiles> directory.
- Verifies successful downloads by comparing sha1sums .
- Unpacks the tarballs into the <installation_directory>/CSPP/SDR_4_1> directory.
- Downloads the CSPP verification test data tarball.

To execute the SDR software, set the CSPP_SDR_HOME environment variable and source the environment variable script. Set the CSPP_SDR_HOME environment variable to the name of the directory where CSPP SDR was installed, and then execute the environment script as shown below:

```
export CSPP_SDR_HOME=<installation_directory>/CSPP/SDR_4_1
source $CSPP_SDR_HOME/cspp_sdr_env.sh
```

Verify your installation by unpacking the test data tarball in a directory separate from the CSPP SDR installation.

```
tar xf <<installation_directory>>/tarfiles/CSPP_SDR_V4.1.1_TEST_DATA.tar.gz
```

This will create a new `cspp_sdr_test_data` directory containing verification datasets.

Proceed to **Sections [3.3](#) and [3.4](#)** to initialize the Look-Up-Tables (LUT) and ancillary data.

3.1.2 Manual Installation

To manually install the software, download the following files from the CSPP website:

<https://cimss.ssec.wisc.edu/cspp/>

```
CSPP_SDR_V4.1.tar.gz
CSPP_SDR_V4.1_static_luts_npp.tar.gz
CSPP_SDR_V4.1_static_luts_j01.tar.gz
CSPP_SDR_V4.1_static_luts_j02.tar.gz
CSPP_SDR_V4.1_straylight_luts_npp.tar.gz
CSPP_SDR_V4.1_straylight_luts_j01.tar.gz
CSPP_SDR_V4.1_straylight_luts_j02.tar.gz
CSPP_SDR_V4.1_static_tiles.tar.gz
CSPP_SDR_V4.1.1_patch.tar.gz
```

Install the software as shown below (a new directory named `SDR_4_1` will be created). In this example, the tar files are assumed to be in the user's home directory.

```
mkdir CSPP
cd CSPP
tar xf ../CSPP_SDR_V4.1.tar.gz
tar xf ../CSPP_SDR_V4.1_static_luts_npp.tar.gz
tar xf ../CSPP_SDR_V4.1_static_luts_j01.tar.gz
tar xf ../CSPP_SDR_V4.1_static_luts_j02.tar.gz
tar xf ../CSPP_SDR_V4.1_static_tiles.tar.gz
tar xf ../CSPP_SDR_V4.1_straylight_luts_npp.tar.gz
tar xf ../CSPP_SDR_V4.1_straylight_luts_j01.tar.gz
tar xf ../CSPP_SDR_V4.1_straylight_luts_j02.tar.gz
tar xf ../CSPP_SDR_V4.1.1_patch.tar.gz
```

Set the `CSPP_SDR_HOME` environment variable to the name of the directory where CSPP SDR was installed (`$HOME` in this example), and then execute the environment script as shown below:

```
export CSPP_SDR_HOME=$HOME/CSPP/SDR_4_1
source $CSPP_SDR_HOME/cspp_sdr_env.sh
```

To verify your installation, download the following file containing the test input and output data:

```
CSPP_SDR_V4.1.1_TEST_DATA.tar.gz
```

The test data should be unpacked in a directory separate from the CSPP SDR installation, e.g.,

```
cd $HOME
tar xf CSPP_SDR_V4.1.1_TEST_DATA.tar.gz
```

This will create a new `cspp_sdr_test_data` directory containing verification datasets.

3.2 Applying the CSPP SDR 4.1.1 Patch to an Existing SDR 4.1 Installation

Download the patch from the CSPP website:

<https://cimss.ssec.wisc.edu/cspp/>

CSPP_SDR_V4.1.1_patch.tar.gz

Set the CSPP_SDR_HOME environment variable to the name of the directory where CSPP SDR was installed (\$HOME in this example), and then execute the environment script as shown below:

```
export CSPP_SDR_HOME=$HOME/CSPP/SDR_4_1
source $CSPP_SDR_HOME/cspp_sdr_env.sh
```

Now apply the CSPP Version 4.1.1 Patch on top of your current installation.

```
cd $CSPP_SDR_HOME
cd ..
tar xf $HOME/CSPP_SDR_V4.1.1_patch.tar.gz
```

Confirm a successfully patched installation by confirming the version number 4.1.1 in the SDR_4_1/RELEASE_NOTES.txt file.

3.3 Updating Look-Up-Tables (LUTs)

After installing CSPP SDR Version 4.1, integrate the VIIRS, ATMS, CrIS and OMPS Look-Up-Tables (LUTs) into your local system database using the sdr_luts.sh script. An internet connection is required for this process. For offline installation, please see [Appendix 1.3: How Can I Execute the CSPP SDR Software without an Internet Connection?](#)

The first time sdr_luts.sh is executed, it creates and populates the local database using the initial set of instrument LUTs that are provided in the “static_luts” tarball. The script also downloads and inserts the most recent LUTs available from the CSPP SDR external distribution site:

http://jpssdb.ssec.wisc.edu/cspp_luts_v70/ ,

into the database. The CSPP SDR processing scripts will fail if sdr_luts.sh is not executed at least once before attempting to create output SDRs.

To download and install the latest LUT files into the local database, run the update script as shown below:

```
sdr_luts.sh
```

This script will create a temporary work directory during execution and remove it upon completion.

By default, `sdr_luts.sh` will only download the most recent LUT files. For historical data processing, use the command `sdr_luts.sh -t <YYYYMMDD>` to download and populate the database for a specific date. For example,

```
sdr_luts.sh -t 20240911
```

will download and populate the LUT database for a VIIRS, CrIS, ATMS or OMPS dataset acquired on 11 September 2024.

Use `-W` to designate a work directory and `-d` to prevent clean-up, for example:

```
sdr_luts.sh -d -W <path to work directory>
```

When new LUT files are posted to the CSPP LUT/ancillary distribution website, they will be downloaded and installed by `sdr_luts.sh`. Run this script weekly to ensure accurate SDR product calibration and navigation. You may want to set up a scheduled execution via the unix “cron” scheduler to routinely check for updated files.

3.4 Updating Dynamic Ancillary Data

The CSPP SDR software requires dynamic ancillary data matching the date and time of the RDR file(s) within a specified tolerance, so it is necessary to update these data files periodically. The CSPP SDR execution scripts will automatically determine the files that are required based upon the RDR(s) date and time range. The software first checks the local cache directory to see if there are any matching Two Line Element (TLE), Polar Wander (PW) and VIIRS-RSBAUTOCAL-HISTORY-AUX files. If no matching files are found, the SDR execution scripts will download the files from the CSPP ancillary data archive:

http://jpssdb.ssec.wisc.edu/cspp_v_4_0/ancillary .

Alternatively, use the dynamic ancillary data update script to download and install the latest ancillary data files (an Internet connection is required):

```
sdr_ancillary.sh
```

This script will create a temporary work directory during execution and remove it upon completion.

Use this script as the optimal method for updating ancillary data each day. Routine daily SDR ancillary script execution (for example using the “cron” scheduler) will ensure the best SDR processing latency. If you do not wish to query the CSPP SDR ancillary external archive for files at runtime as part of the CSPP SDR processing, use the `-l` option in the SDR processing scripts, (e.g., `viirs_sdr.sh -l`).

Use `-W` to designate a work directory and `-d` to prevent clean-up, for example:

```
sdr_ancillary.sh -d -W <path to work directory>
```

By default, `sdr_ancillary.sh` will only download the most recent ancillary files. For historical data processing, use the command `sdr_ancillary.sh -t <YYYYMMDD>` to populate the ancillary database for a specific date. For example,

```
sdr_ancillary.sh -t 20240911
```

will download and populate the ancillary database for a VIIRS, CrIS, ATMS or OMPS dataset acquired on 11 September 2024.

Section 4: VIIRS SDR Software

4.1 VIIRS SDR Driver Script

The principal script is `SDR_4_1/bin/viirs_sdr.sh`, which operates by default in the current directory. It creates intermediate files and soft links, and final HDF5 (*.h5) output files. All files should be removed from the work directory before each execution of `viirs_sdr.sh`. To execute the VIIRS SDR processing script, provide the path and name of the VIIRS RDR file or files to be processed, e.g.

```
viirs_sdr.sh RNSCA-RVIRS_j01_d20240911_t1955034_e2006262_b00001_c20240911200637112000_dr1u_ops.h5
```

or

```
viirs_sdr.sh /data/noaa21/RNSCA-RVIRS_j02*.h5
```

Several options are available for `viirs_sdr.sh` as shown below:

-p <num_procs> Use <num_procs> number of processor cores, -a Aggregate output files into one file per product type (default is no aggregation), -z Compress HDF5 output files with transparent internal gzip (default is no compression), -t N N Trim granules. -t "Trim Begin" "Trim End" -t 1 1 Trims first and last granule, -d Retain all intermediate files created during processing (default is to delete these files), -g Retain ellipsoid geolocation products (GMODO and GIMGO files), -l Do not download ancillary files upon execution (default is to download files if needed), -h Print helpful information. Use this to display all available options.

Examples:

```
viirs_sdr.sh -p 4 /data/RNSCA-RVIRS_*.h5
```

will process 4 VIIRS SDR granules at a time from all the RDRs that are found in the `/data` directory by using 4 processor cores simultaneously. The output files will be identical no matter how many processor cores are used.

```
viirs_sdr.sh -g /data/j02/RNSCA-RVIRS_j02*.h5
```

will create NOAA-21 VIIRS SDR h5 files from all RDR files found in the `/data/j02` directory, and include the creation of both Terrain Corrected (GITCO and GMTCO) and Ellipsoid (GIMGO and GMODO) geolocation files.

```
viirs_sdr.sh -l -a -z -p 2 RNSCA-RVIRS_npp_d20240928_t1715045_e1726273_*.h5
```

will aggregate and internally compress all HDF5 output files, and execute using only local dynamic ancillary files. Two processor cores will be used for creating SDRs and for aggregating the granules.

4.2 Running the VIIRS SDR Test Case

To run the VIIRS NOAA-20 SDR test case, unpack the test data as shown in [Section 3.1](#) and then execute the commands below. Start with the `sdr_luts.sh` command first to ensure that you have the correct Look-Up-Tables for the test data date and time:

```
sdr_luts.sh -t 20240911
cd cspp_sdr_test_data/viirs_test
mkdir work_j01
cd work_j01
viirs_sdr.sh -p 4 ../input_j01/RNSCA-RVIRS_j01*.h5
```

The test case will use 4 processing cores to create 8 output NOAA-20 VIIRS granules, each of 85.7 seconds duration. The processing should be completed in 3-6 minutes on modern computers, resulting in 26 HDF5 files (*_cspp_dev.h5) per granule. The VIIRS processing script will return a status code of zero if it runs normally, and a non-zero code if it encounters a fatal error.

To verify your output files against the output files created by NOAA operations, execute the following commands:

```
cd ..
./viirs_sdr_compare.bash idps_output_j01/ work_j01/
```

This script compares CSPP SDR I-Band and M-Band output file Radiances with operational NOAA IDPS output files for all granules. The number of differences found for each band and granule will be printed. There should be few, if any, differences.

To compare the geolocation file latitude and longitudes, execute this command in the same directory:

```
./viirs_geo_compare.bash idps_output_j01/ work_j01/
```

This script compares the latitude values in the Day/Night Band SDR files (GNDBO), the I-Band geolocation files (GITCO) and the M-Band geolocation files (GMTCO) produced by the CSPP software to the operational NOAA IDPS output files for all granules. Any value that is outside a tolerance of 10^{-4} degrees is printed. That represents about 10 meters at mid-latitudes. There should be few if any differences.

4.3 Creating VIIRS Quicklook Images

This section describes using the image creation script (`ql_sdr.sh`) that is distributed as part of CSPP SDR to quickly create low resolution VIIRS band images. To create high quality, high resolution images, please use the CSPP Polar2Grid software which is distributed through the CSPP website. Documentation for the most recent version of Polar2Grid is available online at:

<https://www.ssec.wisc.edu/software/polar2grid/>.

To create quicklook images of the VIIRS SDRs using the `ql_sdr.sh` script, use this command structure:

```
$CSPP_SDR_HOME/bin/ql_sdr.sh path_to_sdrs/
```

This script will produce an image of each VIIRS band (except the Day/Night Band) using all of the M-band and I-band SDR files found in the `path_to_sdrs/` directory. Using the “`--plot scatter`” option will create images with the lowest latencies. The images will be created in the current directory, for example,

```
ql_sdr.sh --plot scatter work_j01/
```

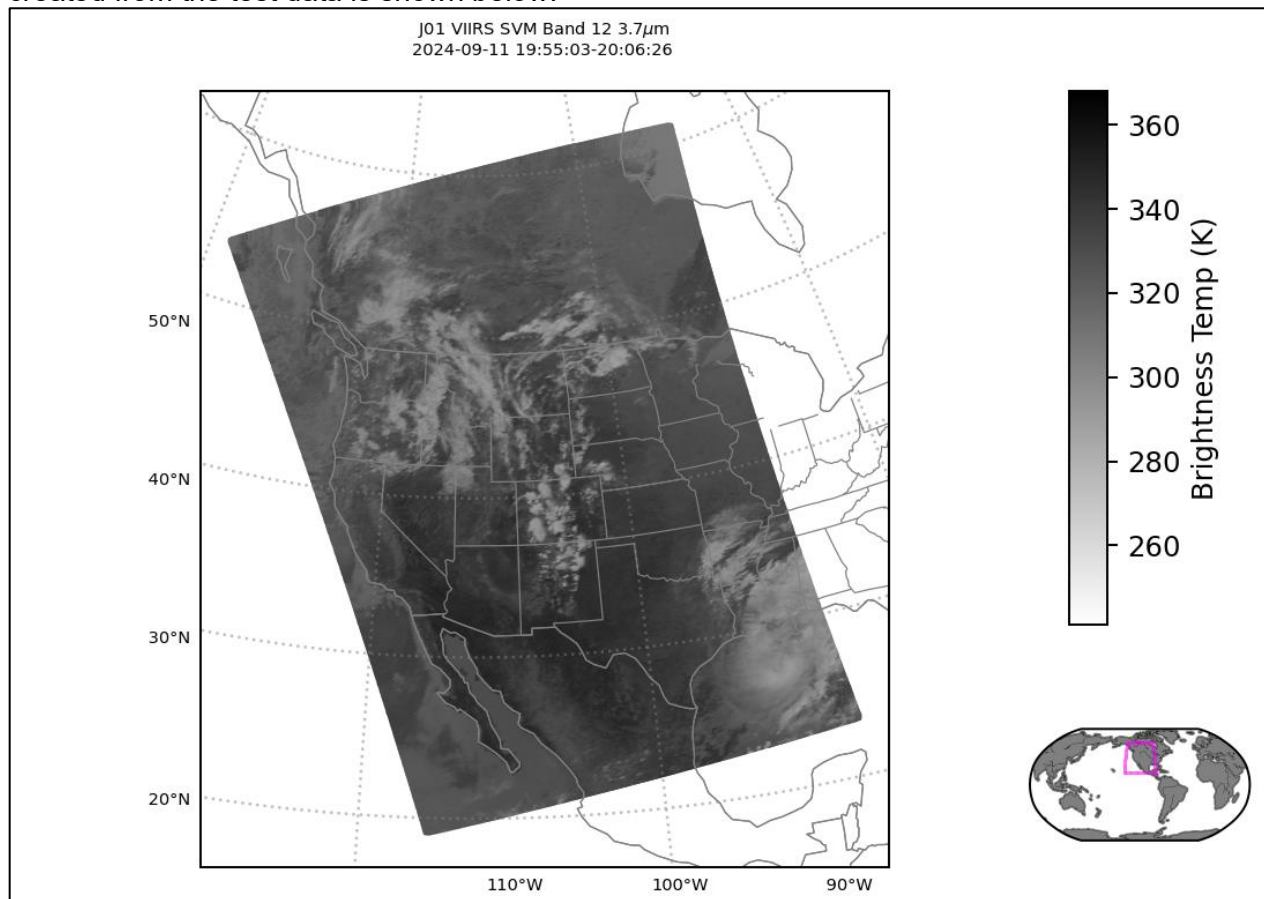
will create images of all the I-Band and M-Band files in the work_j01/ directory and store them in the current directory. You can also create individual band images by using wildcards, for example,

```
ql_sdr.sh --plot scatter work_j01/SVM12*.h5
```

will create an aggregated PNG image of the VIIRS M-Band 12 ($3.7\ \mu\text{m}$) brightness temperatures from all of the SDR granules located in the work directory, as long as the matching geolocation files are collocated. An example of the SVM12 quicklook image,

SVM12_j01_d20240911_t1955034_e1956279_b35320_c20241009205127523555_cspp_dev.png

created from the test data is shown below.



Other quicklook options are available. Use `ql_sdr.sh -h` for a complete list.

Section 5: ATMS SDR Software

5.1 ATMS SDR Driver Script

The principal script is `SDR_4_1/bin/atms_sdr.sh`, which operates by default in the current directory. It creates intermediate files and soft links, and final HDF5 (*.h5) output files. All files should be removed

from the work directory before each execution of `atms_sdr.sh`. There is a cross-granule requirement for calibration of ATMS. Three contiguous good quality science RDR granules must be available to create one output SDR granule.

To execute the ATMS SDR processing script, provide the path and name of the ATMS RDR file or files to be processed, e.g.

```
atms_sdr.sh RATMS-RNSCA_j01_d20240911_t1955160_e2006279_b00001_c20240911200637696000_dr1u_ops.h5
```

or

```
atms_sdr.sh /data/j01/RATMS-RNSCA*.h5
```

Several options are available for `atms_sdr.sh` as shown below:

-p <code><num_procs></code> Use <code><num_procs></code> number of processor cores, -a Aggregate output files into one file per product type (default is no aggregation), -z Compress HDF5 output files with transparent internal gzip (default is no compression), -t N N Trim granules. -t "Trim Begin" "Trim End" -t 1 1 Trims first and last granule, -d Retain all intermediate files created during processing (default is to delete these files), -l Do not download ancillary files upon execution (default is to download files if needed), -s Do not apply science RDR filtering. Caution should be exercised when using this option as it may result in calibration that does not match NOAA operational products, -h Print helpful information. Use this to display all available options.

Examples:

```
atms_sdr.sh -p 2 -a -z RATMS-RNSCA_j01_d20240920_t 1708533_e1709249*.h5
```

will use 2 processor cores to generate the SDRs from the input NOAA-20 RDR, and aggregate and internally compress all HDF5 output files.

```
atms_sdr.sh -p 4 -t 2 1 /RDR/RATMS-RNSCA_j02_d20240814_t2259103_*.h5
```

Executing this command will create NOAA-21 ATMS SDR files from all granules contained in the RDR except the first two, and the last one in the contiguous sequence, using 4 processor cores.

5.2 Running the ATMS SDR Test Case

To run the ATMS NOAA-20 SDR test case, unpack the test data as shown in [Section 3.1](#) and then execute the commands below. Start with the `sdr_luts.sh` command first to ensure that you have the correct Look-Up-Tables for the test data date and time:

```
sdr_luts.sh -t 20260323
cd cspp_sdr_test_data/atms_test
mkdir work_j01
cd work_j01
atms_sdr.sh -p 4 ../input_j01/RATMS-RNSCA_j01*.h5
```

The test case will use 4 processing cores to create 9 output NOAA-20 ATMS granules in less than one minute. The input Science RDR arrays are scanned for size; in this NOAA-20 test case 2 arrays are rejected because they do not meet size thresholds.

For each granule, 3 HDF5 files will be produced (GATMO*.h5, SATMS*.h5 and TATMS*.h5). The ATMS processing script will return a status code of zero if it runs normally, and a non-zero code if it encounters a fatal error.

The ATMS SDR processing operates on input granules of 12 scan-lines each. The “ADL_Unpacker” is used to de-aggregate and extract processing granules from input RDR HDF5 files. For a given contiguous sequence of N raw data record (RDR) processing granules, the user can expect up to N-2 SDR granules, also of 12 scan-lines, will be produced. The first 12 scan-line processing granule and the last 12 scan-line processing granules are used for context and are not processed by the SDR algorithm. This yield may be reduced if there are incomplete intermediate granules.

To verify your output files against the output files created at NOAA operations, execute the following commands:

```
cd ..  
./atms_sdr_compare.bash idps_output_j01/ work_j01/
```

This script compares CSPP ATMS output file Brightness Temperatures and Geolocation file Latitude and Longitudes with operational NOAA IDPS output files for all granules. The number of differences found will be printed. A successful test will result in few, if any differences.

5.3 Creating ATMS Quicklook Images

To create quicklook images of all channels of the ATMS SDRs, use the ql_sdr.sh script:

```
$CSPP_SDR_HOME/bin/ql_sdr.sh path_to_sdrs/
```

This script will produce an image of each ATMS channel (22 bands) from the SATMS*.h5 antenna temperature and GATMO*.h5 geolocation files found in the path_to_sdrs/ directory. The output files will be created in the current directory, e.g.

```
ql_sdr.sh work_j01/
```

will create NOAA-20 ATMS antenna temperature images from files located in the work_j01/ directory and place them in the current directory.

You can also create individual band images by using the -c <channel number (1-22) > option, e.g.,

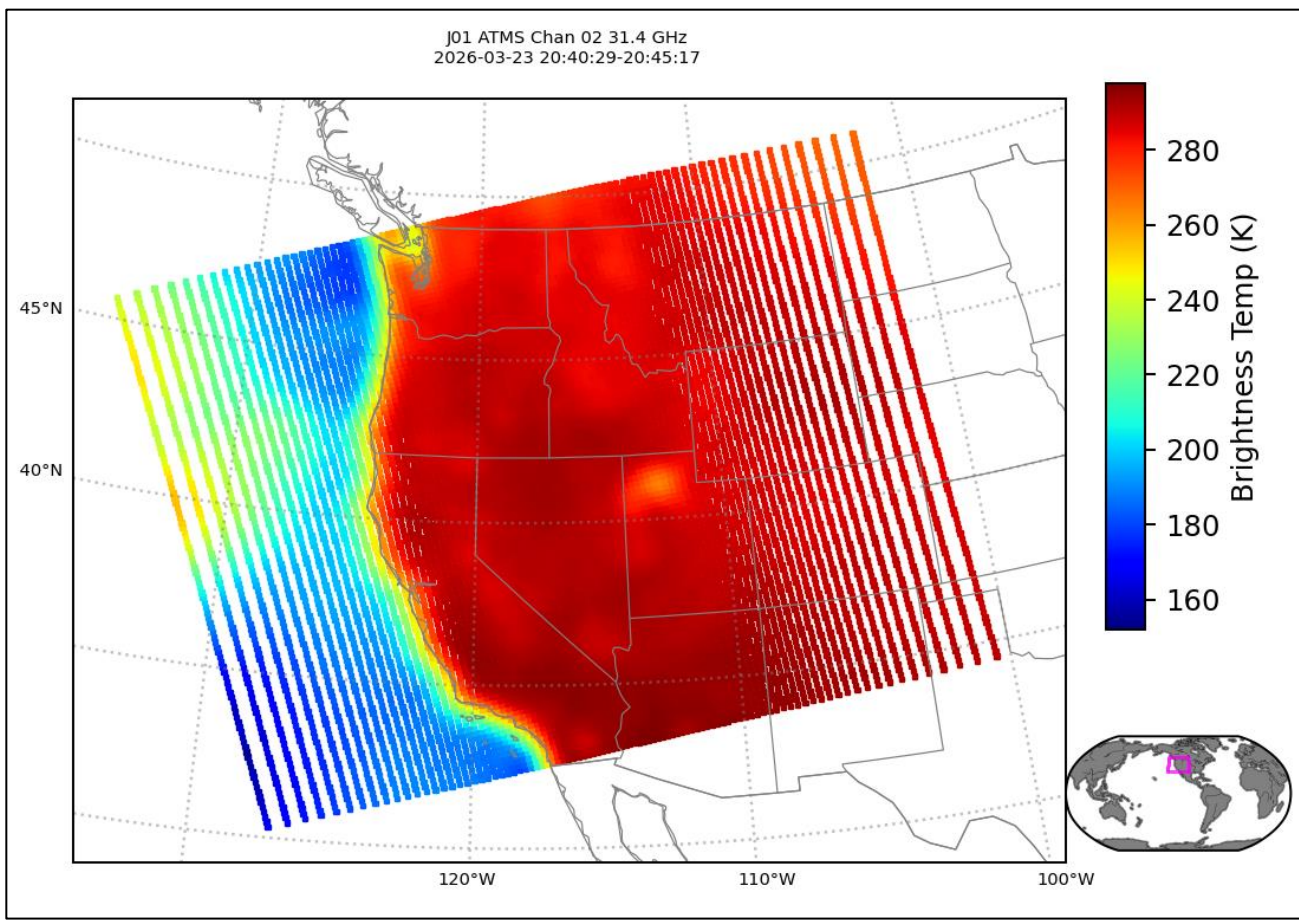
```
ql_sdr.sh -c 2 work_j01/
```

will create an aggregated PNG image of the ATMS Channel 2 antenna temperatures from all of the SDR granules located in the work directory, as long as the matching geolocation files are collocated.

An example of the NOAA-20 ATMS channel 2, 31.4 GHz quicklook image,

SATMS_2_j01_d20260323_t2040296_e2041013_b43237_c20260324205737174861_cspp_dev.png

from the test data set is shown below:



Other quick look options are available. Use `ql_sdr.sh -h` for a complete list.

Section 6: CrIS SDR Software

6.1 CrIS SDR Driver Script

The principal script is `SDR_4_1/bin/cris_sdr.sh`, which operates by default in the current directory. It creates intermediate files and soft links, and final HDF5 (*.h5) output files. All files should be removed from the work directory before each execution of `cris_sdr.sh`. There is a cross-granule requirement for calibration of CrIS; Nine contiguous good quality science RDR granules must be available to create one output SDR granule.

To execute the VIIRS SDR processing script, provide the path and name of the VIIRS RDR file or files to be processed, e.g.

An anomaly occurred on the S-NPP CrIS instrument on 21 May 2021 that ultimately led to loss of the Mid Wave Band. The Mid Wave Band is no longer available from S-NPP. This is discussed in more detail in [Section 2.2.5](#).

To execute the CrIS SDR processing script, provide the path and name of the CrIS RDR file or files to be processed, e.g.

```
cris_sdr.sh RCRIS-RNSCA_j01_d20240911_t1955160_e2006279_b00001_c20240911200637974000_dr1u_ops.h5
```

or

```
cris_sdr.sh /data/j02/RCRIS-RNSCA_j02*.h5
```

Several options are available for `cris_sdr.sh` as shown below:

-p <num_procs> Use <num_procs> number of processor cores, -a Aggregate output files into one file per product type (default is no aggregation), -z Compress HDF5 output files with transparent internal gzip (default is no compression), -t N N Trim granules. -t "Trim Begin" "Trim End" -t 1 1 Trims first and last granule, -d Retain all intermediate files created during processing (default is to delete these files), -l Do not download ancillary files upon execution (default is to download files if needed), -s Do not apply science RDR filtering. Caution should be exercised when using this option as it may result in calibration that does not match NOAA operational products, -h Print helpful information. Use this to display all available options.

Examples:

```
cris_sdr.sh -p 4 /data/RCRIS-RNSCA_j01*.h5
```

will process 4 NOAA-20 CrIS SDR granules at a time from all the RDRs that are found in the /data directory using 4 processor cores and produce Full Spectral Resolution (FSR) output files.

```
cris_sdr.sh -l -a -z RCRIS-RNSCA_j02_d20240817_t1743284_e1755443.h5
```

will aggregate and internally compress all NOAA-21 CrIS Full Spectral Resolution (FSR) HDF5 output files, and execute using only local dynamic ancillary files.

6.2 Running the CrIS SDR Test Case

To run the CrIS NOAA-20 SDR Full Spectral Resolution (FSR) test case, unpack the test data as shown in [Section 3.1](#) and then execute the commands below. Start with the `sdr_luts.sh` command first to ensure that you have the correct Look-Up-Tables for the test data date and time:

```
sdr_luts.sh -t 20240911
cd cspp_sdr_test_data/cris_test
mkdir work_fsr_j01
cd work_fsr_j01
cris_sdr.sh -p 4 ../input_j01/RCRIS-RNSCA_j01*.h5
```

The test case executed using 4 processing cores should run in 2-4 minutes and create 12 output file HDF5 granules. The input Science RDR arrays are scanned for size; in this NOAA-20 test case 1 array is rejected because they do not meet size thresholds.

For each granule, 2 HDF5 files are produced (SCRIF* and GCRSO* files). To improve processing latency, upfront calculations are done only once for each processor core (4 for our test case). If the CrIS processing script runs normally, it will return a status code equal to zero. If the CrIS processing script encounters a fatal error, it will return a non-zero status code.

Note that the cross-granule requirements for the CrIS SDR are 4 "before" granules and 4 "after" granules. Therefore, a sequence of 9 consecutive granules is required to create a single granule to SDR. For example, if the script is called on a single RDR file containing 15 consecutive granules, the middle 7 granules will be processed to SDRs. If the script is executed on two RDR files containing 30 consecutive granules, the middle 22 granules will be processed. If for any reason there is an incomplete or corrupted granule in the sequence, the count of 4 granules for context needs to start again.

To verify your output FSR files against the output CrIS SDRs created at NOAA operations, execute the following commands:

```
cd ..  
./cris_sdr_fsr_compare.bash idps_output_j01/ work_fsr_j01/
```

This script compares CSPP CrIS NOAA-20 FSR output file arrays and Geolocation file Latitude and Longitudes with operational NOAA IDPS output files for all granules. The number of differences found will be printed. A successful test will result in few, if any differences.

6.3 Creating CrIS Quicklook Images

To create quicklook images of the CrIS FSR Shortwave, Midwave (excluding S-NPP) and Longwave infrared band brightness temperatures, use the `ql_sdr.sh` script:

```
$CSPP_SDR_HOME/bin/ql_sdr.sh path_to_sdrs/
```

This script will produce images of brightness temperatures for the infrared Longwave, Midwave (NOAA-20 and NOAA-21 only) and Shortwave bands using the `SCRIF*.h5` radiance and `GCRSO*.h5` geolocation files in the `path_to_sdrs/` directory.

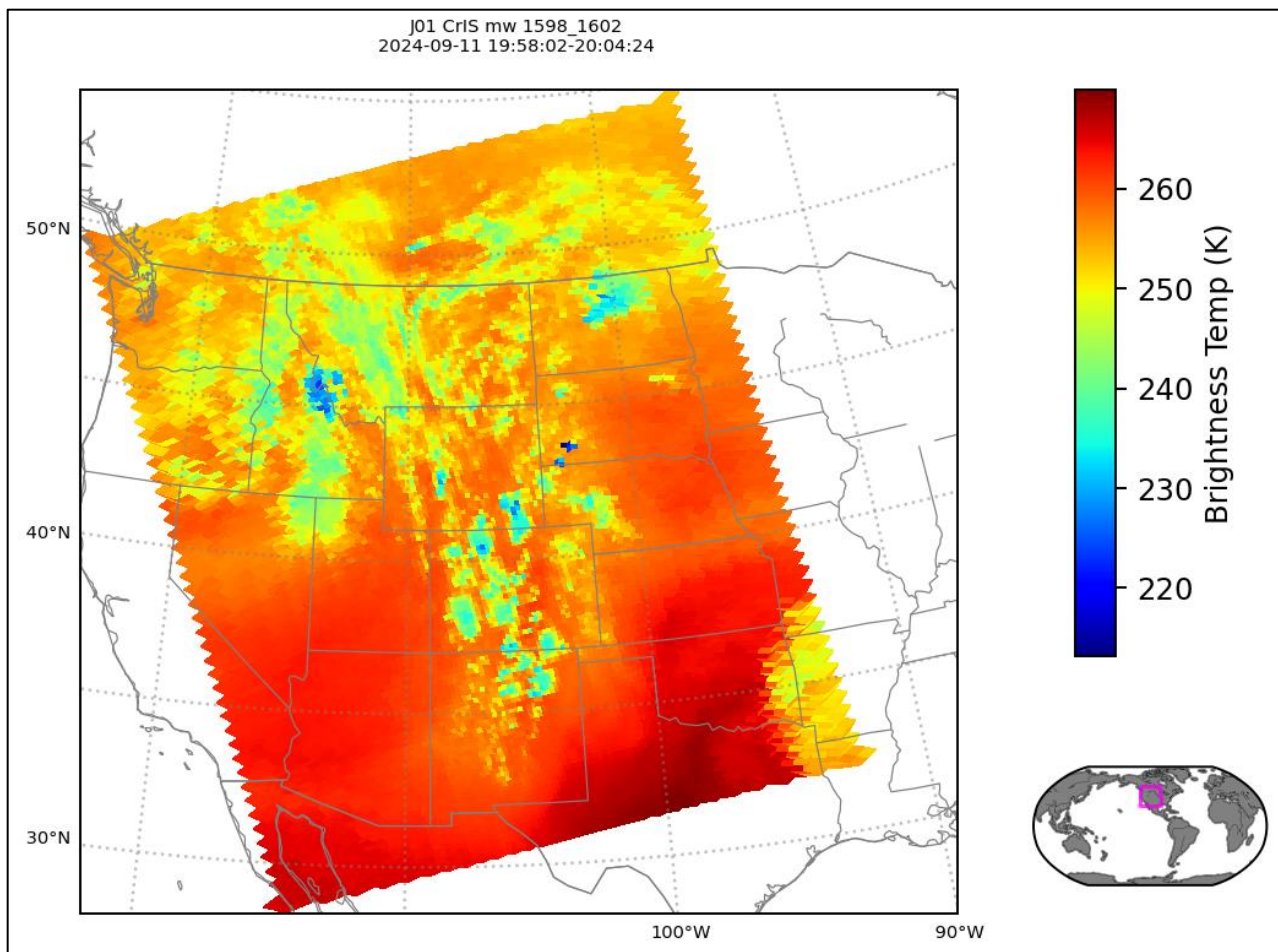
The output files will be created in the execution directory, e.g.

```
ql_sdr.sh --plot pcolormesh work_fsr_j01/
```

will create brightness temperature images from files located in the `work_fsr_j01/` directory and place them in the current directory. An example of the quicklook CrIS infrared Midwave Band image,

`SCRIF_2_j01_d20240911_t1957309_e1958007_b35320_c20241009222933811404_cspp_dev.png`

created from the NOAA-20 CrIS test data is shown below.



Other quick look options are available. Use `ql_sdr.sh -h` for a complete list.

Section 7: OMPS SDR Software

7.1 OMPS SDR Driver Scripts

The Ozone Mapping and Profiler Suite (OMPS) comprises three instruments: the Nadir Mapper (NM), also referred to as the Nadir Total Column (TC) instrument; the Nadir Profiler (NP); and the Limb Profiler (LP). The CSPP SDR software generates products for the Nadir Mapper and Nadir Profiler, but not for the Limb Profiler, as ADL does not support it.

The principal script is `SDR_4_1/bin/omps_sdr.sh`, which operates by default in the current directory. The script can be used to create either Nadir Mapper or Nadir Profiler SDRs depending on the input RDR(s) provided. It creates intermediate files and soft links, and final HDF5 (*.h5) output files. All files should be removed from the work directory before each execution of `omps_sdr.sh`.

OMPS instrument characteristics differ with satellite. Table 1 contains details on the resolution of the Nadir Mapper and Nadir Profiler instruments onboard the NOAA-21, NOAA-20 and S-NPP platforms. Although the product output SDR HDF5 array sizes for each instrument are the same, the number of valid data in the files differs with the number of FOVs shown in the table below. If the valid data does not fill the array, the remainder of the array cells are padded with Fill Values.

Table 1: OMPS Nadir Mapper (NM) and Nadir Profiler (NP) instrument characteristics for each JPSS satellite, including spatial resolution and Fields-of-View (FOV) per granule.

Satellite	Sensor	FOV (Along Track x Cross Track)	Nadir Horizontal Resolution	Sensor	FOV	Nadir Horizontal Resolution
Suomi-NPP	NM	5 x 35	50 x 50km	NP	1 x 1	250 x 250 km
NOAA-20	NM	15 x 35	50 x 17 km	NP	5 x 5	50 x 50 km
NOAA-21	NM	30 x 177	12 x 10 km	NP	5 x 5	50 x 50 km

To execute the OMPS SDR processing script, you must provide the path and name of the OMPS RDR file or files to be processed, e.g.

```
omps_sdr.sh RNSCA-ROTCS_j01_d20240911_t1955175_e2006308_b00001_c20240911200637767000_dr1u_ops.h5
```

```
omps_sdr.sh /data/j02/RNSCA-RONPS_j02*.h5
```

One or more RDR files in sequential time order from the same instrument and satellite can be processed in one execution. The script will not process both NP and NM RDR files at the same time even if they are from the same instrument. Separate calls for each instrument are required.

Several options are available for `omps_sdr.sh` as shown below:

- p** <num_procs> Use <num_procs> number of processor cores,
- a** Aggregate output files into one file per product type (default is no aggregation),
- z** Compress HDF5 output files with transparent internal gzip (default is no compression),
- t N N** Trim granules. -t "Trim Begin" "Trim End" -t 1 1 Trims first and last granule,
- d** Retain all intermediate files created during processing (default is to delete these files),
- l** Do not download ancillary files upon execution (default is to download files if needed),
- h** Print helpful information. Use this to display all available options.

Examples:

```
omps_sdr.sh -p 4 \
RNSCA-RONPS_j01_d20240723_t1828073_e1841128_b00001_c20240723184056526000_all-_dev.h5
```

This command will create OMPS Nadir Profiler SDRs using 4 processing cores from a single NOAA-20 input RDR file.

```
omps_sdr.sh -d -l -a /data/RNSCA-ROTCS_j02_d20240623*.h5
```

This command will aggregate and internally compress all NOAA-21 OMPS Nadir Mapper HDF5 output files, and execute using only local dynamic ancillary files. The script will process all sequential Nadir Mapper RDR files found in the /data directory from 23 June 2024.

7.2 Running the OMPS SDR Test Cases

To run the OMPS NOAA-20 SDR test cases, unpack the test data as shown in [Section 3.1](#) and then execute the commands below. Start with the `sdr_luts.sh` command first to ensure that you have the correct Look-Up-Tables for the test data date and time:

```
sdr_luts.sh -t 20240911
cd cspp_sdr_test_data/omps_test
mkdir work_j01
cd work_j01
omps_sdr.sh -p 4 ../input_j01/RNSCA-ROTCS_j01*.h5
omps_sdr.sh -p 4 ../input_j01/RNSCA-RONPS_j01*.h5
```

A single execution is required for creating products for each instrument. RNSCA-ROTCS is the Nadir Mapper RDR prefix, and the RNSCA-RONPS is the Nadir Profiler RDR prefix. The NM test case will use 4 processing cores and run to completion in less than 1 minute and create 16 output granules. The NP test case also will produce 16 output granules and will run to completion in less than 1 minute.

For each Nadir Mapper granule, 2 HDF5 files will be produced (SOMTC*.h5 and GOTCO*.h5 files). For each Nadir Profile granule, 2 HDF5 files will be produced (SOMPS*.h5 and GONPO*.h5 files). If the OMPS processing scripts run normally, they will return a status code equal to zero. If the OMPS processing scripts encounter a fatal error, they will return a non-zero status code.

The OMPS NM and NP processing requires 3 sequential RDR granules in order to produce one output SDR granule. The first and last granules are used for context.

To verify your output files against the output files created at NOAA operations, execute the following commands:

```
cd ..
./omps_sdr_compare.bash idps_output_j01/ work_j01/
```

This script compares OMPS Radiances and geolocation Latitude and Longitudes from both the OMPS NM and NP output files with operational NOAA IDPS output files for all granules. The number of differences found will be printed. A successful test will result in few, if any differences.

7.3 Creating OMPS Quicklook Images

The CSPP quick look software supports a subset of OMPS Nadir Mapper (NM) and Nadir Profiler (NP) SDR bandwidth channels. To execute the quicklook script, use `ql_sdr.sh`:

```
$CSPP_SDR_HOME/bin/ql_sdr.sh path_to_sdrs/
```

This script will create radiance images from 13 NM channels from the SOMTC*.h5 radiance and GOTCO*.h5 geolocation files found in the `path_to_sdrs/` directory. It will also create 7 NP channel images from the SOMPS*.h5 radiance and GONPO*.h5 geolocation files found in the `path_to_sdrs/` directory. One execution will create both sets of image files. The output files will be created in the current directory.

The following is a list of channels and wavelengths that are supported by the CSPP SDR quicklook software for the OMPS Nadir Mapper:

OMPS Nadir Mapper Channel Number	Wavelength
23	308.8 nm
28	310.7 nm
31	312.0 nm
32	312.4 nm
34	313.2 nm
37	314.5 nm
44	317.4 nm
56	322.4 nm
77	331.2 nm
97	339.5 nm
146	360.0 nm
177	372.9 nm
190	378.4 nm

The following is a list of channels and wavelengths that are supported by the CSPP SDR quicklook software for the OMPS Nadir Profiler:

OMPS Nadir Profiler Channel Number	Wavelength
12	308.7 nm
60	310.7 nm
82	312.0 nm
93	312.4 nm
104	313.2 nm
116	314.5 nm
126	317.4 nm

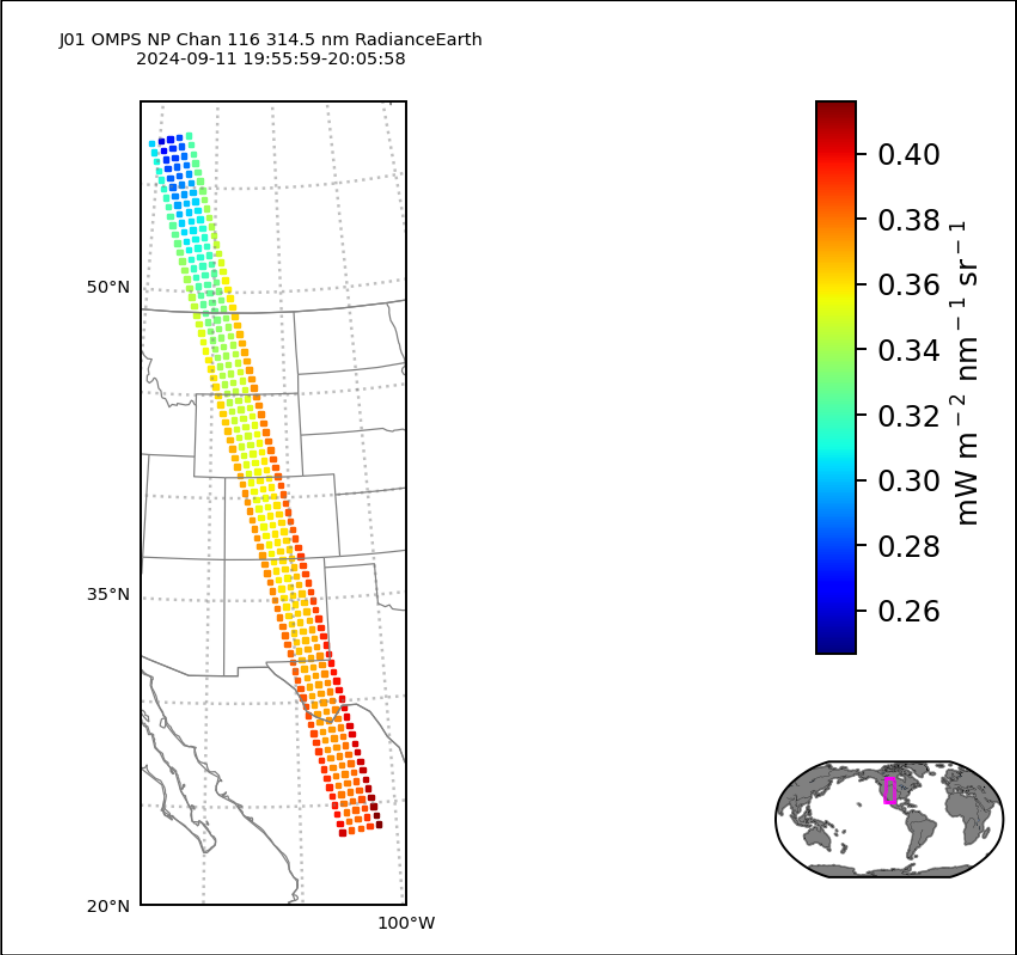
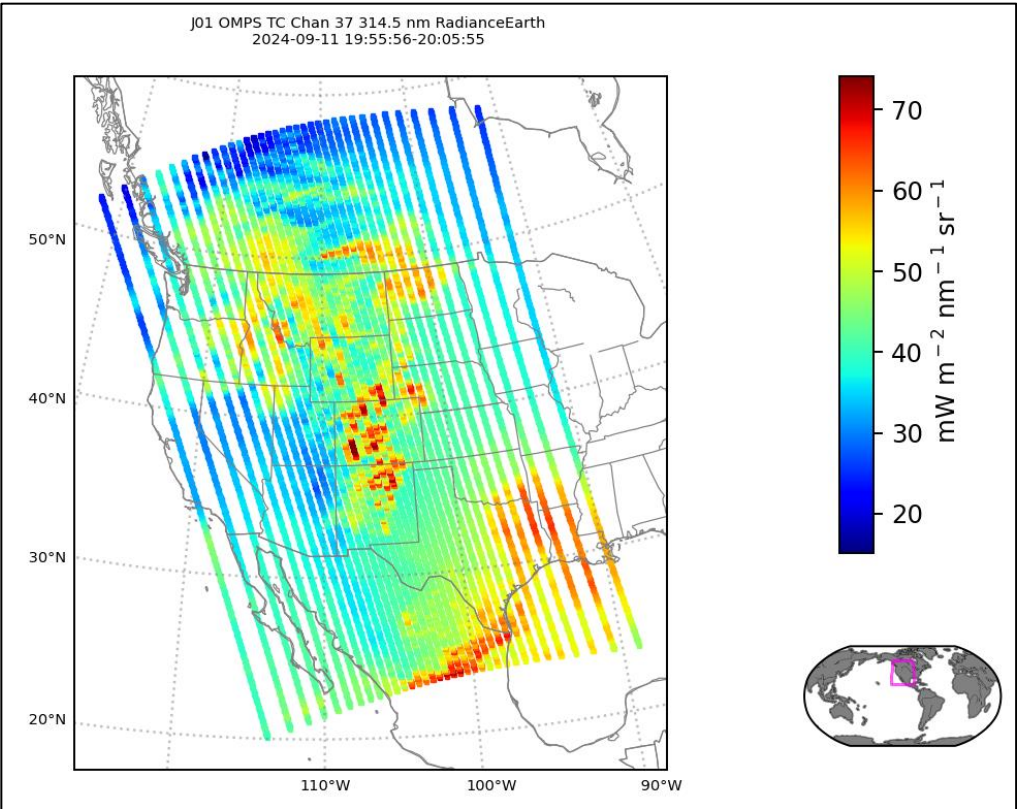
The output files will be created in the execution directory, e.g.

```
q1_sdr.sh work_j01/
```

will create radiance images from files located in the work_j01/ directory and place them in the current directory. In this case, both OMPS NM and OMPS NP images will be created since both sets of files are available in our work_j01 example.

The following example images are made from the NOAA-20 OMPS NM and NP 314.5 nm SDR files (OMPS NM channel 37, and OMPS NP channel 116) test data output located in the work_j01/ directory.

```
SOMTC_37_j01_d20240911_t1955568_e1956342_b35320_c20241010161506671390_cspp_dev.png
SOMPS_116_j01_d20240911_t1955593_e1956367_b35320_c20241010161643940792_cspp_dev.png
```



Appendix 1: Frequently Asked Questions

A1.1 What should I do if I have a problem with CSPP SDR software?

If you experience problems when executing the CSPP SDR software, please follow these steps:

1. Make sure that you read the documentation and follow the instructions as written.
2. Run the verification tests to ensure a successful installation.
3. If you are still having problems, please send an email to cspp_issues@ssec.wisc.edu with a detailed description of the issue. Please include the following information,
 - a. Exact CSPP SDR command you are using;
 - b. Computer hardware specifications, including memory and hard drive space;
 - c. Operating system name and version number;
 - d. If the CSPP SDR execution failed, please send any log files that remain.

A1.2 How Can I Change the Default Look-Up-Table (LUT) and Ancillary Data Locations?

The CSPP SDR Look-Up-Tables (LUTs) and dynamic Ancillary Data external archive locations are identified using two environmental variables:

```
JPSS_REMOTE_LUT_DIR=http://jpssdb.ssec.wisc.edu/cspp\_luts\_v\_7\_0/  
JPSS_REMOTE_ANC_DIR=http://jpssdb.ssec.wisc.edu/cspp\_v\_4\_0/ancillary
```

Once you have sourced the environment variable script as part of your installation, you can change these default locations using the export command or change the protocol to https. For instance:

```
export JPSS_REMOTE_LUT_DIR=http://my.local.lut_archive/cspp/  
export JPSS_REMOTE_LUT_DIR=https://jpssdb.ssec.wisc.edu/cspp\_luts\_v\_7\_0/
```

The sub-directory naming conventions for the SDR dynamic ancillary data must conform to the structure found on the archive site, which consists of data in daily directories:

```
YYYY_MM_DD_JDAY where,  
  YYYY is the year,  
  MM is the month  
  DD is the day of the month, and  
  JDAY is the Julian Day or Day of the Year (1-366)
```

For instance, CSPP SDR ancillary data for 22 August 2024, can be found in this directory:

http://jpssdb.ssec.wisc.edu/cspp_v_4_0/ancillary/2024_08_22_235/

A1.3 How Can I Execute the Software without an Internet Connection?

CSPP SDR software requires dynamic ancillary data files and Look-Up-Tables (LUTs). An internet connection is necessary in order to download the required files. Some users have requested instructions for installation of CSPP SDR on computers that do not have Internet access. This requires installation of CSPP SDR on two computers:

Computer A does not have an Internet connection, and will be used for CSPP SDR processing;
Computer B does have an Internet connection, and will be used only for LUT and ancillary downloads.

1. On **Computer A**, follow the instructions in [Section 3.1](#) to install CSPP SDR Version 4.1. However, when running the LUT and ancillary update scripts as described in [Sections 3.2](#) and [3.3](#), use the `-l` option to set up the local database without downloading files:

```
sdr_luts.sh -l
sdr_ancillary.sh -l
```

2. On **Computer B**, do another full installation of the CSPP SDR Version 4.1 software following the instructions in Section 3.1. When running the LUT and ancillary update scripts as described in Sections 3.2 and 3.3, use the `-s` option to download new LUT and ancillary files without inserting them into a database:

```
sdr_luts.sh -s
sdr_ancillary.sh -s
```

3. On **Computer B**, copy the new LUT and ancillary data files to **Computer A**. The `rsync` command is recommended for this task as shown below (note that we assume a local network is available between Computer B and Computer A):

```
# Set CSPP SDR destination and path on Computer A
sdr_dest=username@computera
sdr_path=/home/username/CSPP/SDR_4_1

# Copy ancillary files to Computer A
rsync -av ${CSPP_SDR_HOME}/anc/cache/20* $sdr_dest:$sdr_path/anc/cache

# Copy LUTs to Computer A
rsync -av ${CSPP_SDR_HOME}/anc/cache/incoming_luts/* \
    $sdr_dest:$sdr_path/anc/cache/incoming_luts
```

4. On **Computer A**, update the CSPP SDR LUT and ancillary database as shown below:

```
sdr_luts.sh -l
sdr_ancillary.sh -l
```

When you execute the CSPP SDR processing scripts (`viir_sdr.sh`, `cris_sdr.sh`, `atms_sdr.sh` or `omps_sdr.sh`) on **Computer A**, you must use the `-l` option so that the software will not attempt to download LUT and ancillary files. For example,

```
viirs_sdr.sh -l -p 4 ../rdr/RNSCA-RVIRS*.h5
```

Finally, you must set up a routine download and copy of CSPP SDR ancillary and LUT files as shown in steps 3-5 above, preferably via the cron scheduler. Ancillary files (`sdr_ancillary.sh`) should be downloaded and transferred once a day, while LUTs (`sdr_luts.sh`) should be downloaded and transferred once a week.

Appendix 2: Converting SDR to BUFR Format for NWP Assimilation

A2.1 Script to convert ATMS SDR to DBNet BUFR (atms_bufr.bash)

```
#!/bin/bash

# This script converts all ATMS SDR files in the current directory to BUFR using AAPP
# AAPP can be downloaded from the EUMETSAT NWP SAF (https://nwp-saf.eumetsat.int/)
# For more information see the "WMO Guide to DBNet"
# Liam.Gumley@ssec.wisc.edu

# Set up AAPP environment
export AAPP_HOME=$HOME/install # Change this line for your AAPP installation
export AAPP_PREFIX=$AAPP_HOME/AAPP_8.10
source $AAPP_PREFIX/ATOVS_CONF
export LD_LIBRARY_PATH=$AAPP_PREFIX/AAPP/lib:$LD_LIBRARY_PATH
export PATH=$AAPP_PREFIX/AAPP/bin:$AAPP_HOME/eccodes/bin:$PATH

# Set BUFR environment variables
export ORIGINATING_CENTRE=176 # Change this line for your DBNet center
export SUB_CENTRE=30 # Change this line for your DBNet site

# Create ATMS DBNet BUFR file
sdr=(SATMS*.h5)
tdr=(TATMS*.h5)
geo=(GATMO*.h5)
num=${#sdr[*]}
for ((i = 0; i < num; i++)); do atms_sdr -g ${geo[$i]} ${sdr[$i]} ${tdr[$i]}; done
combine_1c SATMS*.11c
eccodes_encodebufr_1c -i atms.11c ATMS
prepare_dbnet_bufr_for_gts atms.bufr
```


A2.2 Script to convert CrIS FSR SDR to DBNet BUFR (cris_bufr.bash)

```
#!/bin/bash

# This script converts all CrIS FSR SDR files in the current directory to BUFR using AAPP
# AAPP can be downloaded from the EUMETSAT NWP SAF (https://nwp-saf.eumetsat.int/)
# For more information see the "WMO Guide to DBNet"
# Liam.Gumley@ssec.wisc.edu

# Set up AAPP environment
export AAPP_HOME=$HOME/install # Change this line for your AAPP installation
export AAPP_PREFIX=$AAPP_HOME/AAPP_8.10
source $AAPP_PREFIX/ATOVIS_CONF
export LD_LIBRARY_PATH=$AAPP_PREFIX/AAPP/lib:$LD_LIBRARY_PATH
export PATH=$AAPP_PREFIX/AAPP/bin:$AAPP_HOME/eccodes/bin:$PATH

# Set BUFR environment variables
export ORIGINATING_CENTRE=176 # Change this line for your DBNet center
export SUB_CENTRE=30 # Change this line for your DBNet site

# Combine CrIS SDR files into one L1C file
sdr=(SCRIF*.h5)
geo=(GCRSO*.h5)
num=${#sdr[*]}
for ((i = 0; i < num; i++)); do cris_sdr -g ${geo[$i]} ${sdr[$i]}; done
combine_1c SCRIF*.11c
cris_channels cris.11c cris_selection.11c
eccodes_encodebufr_1c -i cris_selection.11c CRISFSR
prepare_dbnet_bufr_for_gts cris_selection.bufr
```